

1 · What Is a CTA Strategy

Harry Chen

Contents

1.1	What a CTA Trades	2
1.1.1	Index, ETF, or future	2
1.1.2	Why futures rather than ETFs	3
1.2	Momentum, the Source of the Return	4
1.2.1	What momentum is	4
1.2.2	Why momentum might work	4
1.3	Long and Short, the Two Sides of a Position	4
1.3.1	What the two sides mean	4
1.3.2	How a short is borrowed	5
1.3.3	Why the short leg is the fragile one	5
1.3.4	How a short is represented	6
1.4	Who Else Is Trading	7
	Background	7
	Appendix · Notation	8

1.1 What a CTA Trades

Definition 1.1 (CTA strategy). A *CTA* (*Commodity Trading Advisor*) strategy is a rule-based strategy that trades **futures** systematically across asset classes—equities, fixed income, commodities, currencies—targeting **absolute return**, independent of market direction.

Note. The name is a historical artifact. Modern CTAs are not commodity-only; typical markets are equity-index futures, Treasuries, FX, energy, metals, and agriculture.

1.1.1 Index, ETF, or future

“S&P 500” names three different objects. Only two of them can be traded.

Layer	Example	What it is	Tradeable
Index	SPX, NDX	A published number computed from its constituents	No
ETF	SPY, VOO, QQQ	A fund holding the constituents; its own shares trade	Yes
Future	ES, NQ	A contract to settle at a future date; holds nothing	Yes

Definition 1.2 (Index).

$$I_t = \frac{M_t}{D_t}, \quad M_t = \sum_i N_i P_{it}$$

where P_{it} is the price of constituent i , N_i its float-adjusted share count, M_t the total float capitalisation, and $D_t > 0$ the *divisor*.

Claim 1.3. Index returns are the capitalisation-weighted average of constituent returns; the divisor sets only the level.

Proof. Both halves follow from I_t being a *sum divided by a constant*. A constituent’s dollar move is N_i times its price change, so its share of the index move is its share of total capitalisation:

$$r_I = \sum_i w_{it} r_i, \quad w_{it} = \frac{N_i P_{it}}{M_t}, \quad \sum_i w_{it} = 1.$$

And D_t divides I_t at both dates, so it cancels from the ratio—it can shift the level anywhere without touching a single return. $\square$

Note (Two consequences). **The top names dominate.** At $w \approx 0.07$ against $w \approx 0.0002$, the largest constituent carries roughly **350 times** the index impact of the smallest; this is not an average of 500 *prices*. **And the level is arbitrary.** D_t cancels from returns, so the scale is whatever the base period set it to—for the S&P 500, $1941-43 = 10$. **Compare returns, never levels.**

A price level is meaningful only against that same series' own history. Across tickers it carries no information at all—what each ETF's share price happens to be was fixed by how the fund was structured at inception. Dividing every series by its own first observation removes that arbitrariness:

Ticker	Raw close, start	Raw close, end	Rebased	Return
SPY	325.05	740.83	227.9	+127.9%
TLT	137.10	87.35	63.7	−36.3%
GLD	143.97	368.49	255.9	+155.9%

SPY has the highest closing price on every single day of the sample, yet GLD returned more. The end level alone cannot tell you this; you need the start it is measured against.

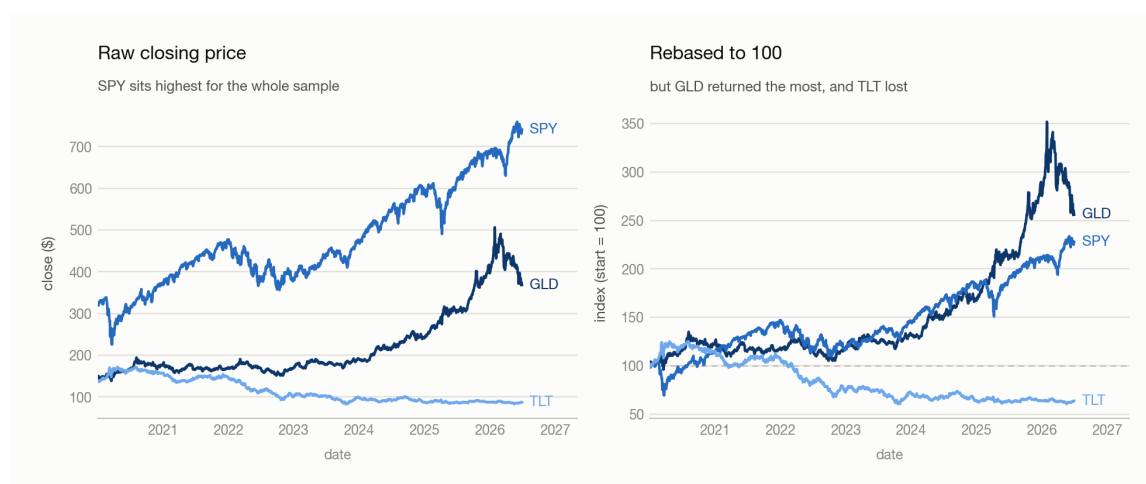


Figure 1.1: The same three ETFs, twice. Left, raw closing prices: SPY runs highest for the entire sample. Right, each series divided by its own first value and multiplied by 100: GLD now finishes highest and TLT ends below the line it started on. The ordering of the levels and the ordering of the returns are different orderings.

1.1.2 Why futures rather than ETFs

Reason	Futures	ETFs
Symmetry	Shorting costs nothing extra	Shorting needs borrowed shares, a fee, availability
Coverage	Deep markets in all four asset classes	Poor in commodities and FX
Capital efficiency	Margin is roughly 5 to 10 percent of notional; spare cash earns col-lateral yield	Full notional tied up

Symmetry is decisive: a strategy that must go short as freely as long cannot tolerate the asymmetry, and §1.3.3 shows why the short leg is the fragile one. The cost is that

futures expire, so a continuous series has to be stitched together from contracts—see [100 · The Dataset](#).

1.2 Momentum, the Source of the Return

1.2.1 What momentum is

Definition 1.4 (Momentum). *Momentum* is the empirical tendency for recent relative performance to persist: assets that outperformed over the past one to twelve months tend to keep outperforming, and recent laggards tend to keep lagging.

Note. The claim is about *returns*, not price levels—it concerns the ordering of performance across assets and across time. That is why one momentum rule can be applied to markets with nothing else in common, from Treasury futures to natural gas.

Turning this into a number—a lookback window, a normalization, a test that it carries information—is [2 · Testing a Signal](#)’s subject. Here it is only the premise.

1.2.2 Why momentum might work

Two contested explanations, no proof. Momentum has been durably profitable regardless.

- **Information diffusion.** Unglamorous macro news spreads gradually, creating sustained pressure. Possibly weakened since 2008, as computing and data distribution got faster.
- **Selective momentum capture.** Winners keep winning and losers keep losing, because large positions take time to build and unwind, investors chase winners, and major events do not resolve in a single day.

[2 · Testing a Signal](#) turns this from a story into something testable.

1.3 Long and Short, the Two Sides of a Position

1.3.1 What the two sides mean

Definition 1.5 (Long). Buy first, sell later. Profit is the sell price minus the buy price.

Definition 1.6 (Short). Borrow and sell first, buy back and return later. Profit is the sell price minus the buy-back price.

Note. Running both deploys capital fully—a 150/50 or 200/100 book rather than idle cash—and lets the strategy express a negative view, which a long-only book cannot.

1.3.2 How a short is borrowed

A short is not selling out of thin air—the shares are borrowed before they are sold:

1. **Borrow** N shares from a lender—a broker, or a long-term holder.
2. **Sell** them now, and receive cash.
3. **Buy back** N shares later, from the market.
4. **Return** the N shares, closing the position.

Note. You never create shares. They are fungible, so returning “ N shares of SPY”—not specific certificates—settles the debt. **Lenders are paid.** Long-term holders earn a **lending fee** on stock they were going to sit on anyway. **And the broker sits in the middle**, matching the two sides and holding margin; the market for this is **securities lending**.

→ The full mechanics beneath this—who holds the shares mid-loan, who receives the dividend, hard-to-borrow rates, and how a short squeeze runs away: §10 of the `market-101-foundations` note in `market_knowledge/`.

1.3.3 Why the short leg is the fragile one

Claim 1.7. For a static position, a long’s loss is bounded by its outlay; a short’s is not.

Proof. With signed share count s , entry price $P_0 > 0$ and exit price P_T , the payoff is

$$\text{PnL}(P_T) = s(P_T - P_0), \quad P_T \in [0, \infty)$$

the domain being half-open because a price cannot go negative.

For a **long**, $s > 0$: PnL is increasing in P_T , so its infimum is at the left endpoint, $\text{PnL} \geq s(0 - P_0) = -sP_0$ —exactly the amount paid. For a **short**, $s < 0$: PnL is decreasing in P_T , and the domain has no right endpoint, so $\text{PnL} \rightarrow -\infty$ as $P_T \rightarrow \infty$. □

The asymmetry is in the **domain**, not the payoff—both are lines of slope $|s|$. The floor exists only because prices are floored at zero.

Note. Hence margin requirements and forced buy-ins—and why a 150/50 book, 200 percent gross, is **two times leverage** rather than free money.

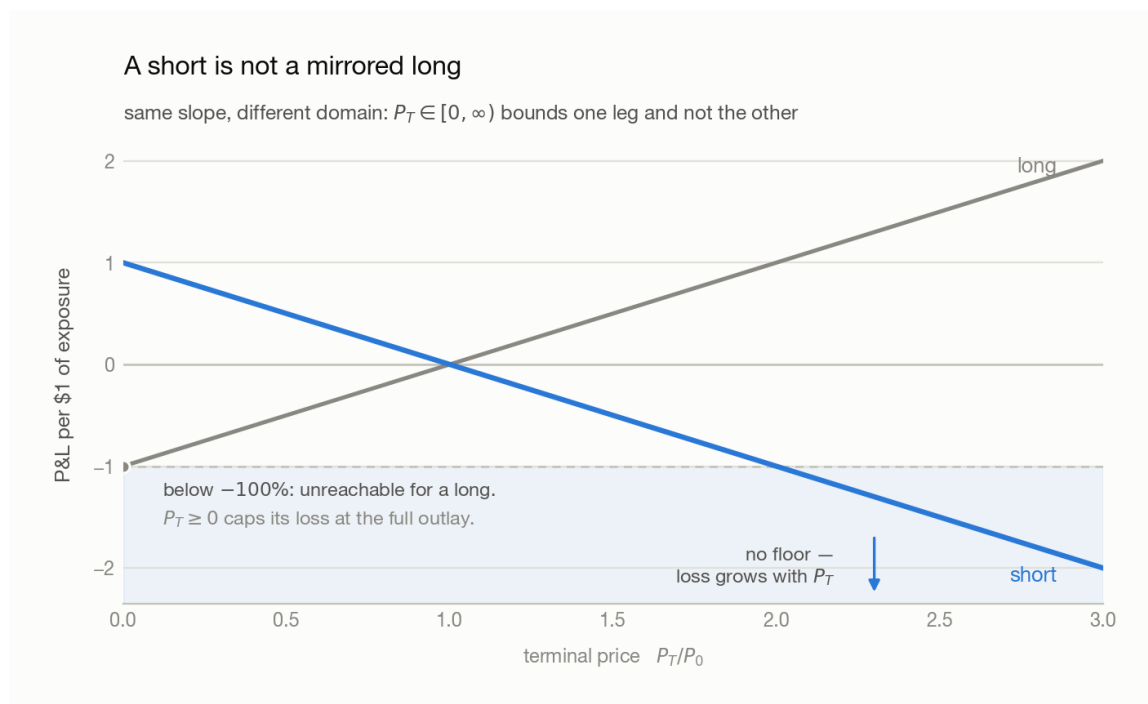


Figure 1.2: Profit and loss per dollar of exposure against the terminal price P_T . The long and short payoffs are lines of equal and opposite slope crossing at the entry price P_0 ; the long's line stops at -100 percent where the price reaches zero, while the short's continues downward without limit as the price rises. Illustrative.

1.3.4 How a short is represented

A short needs no separate machinery. Carry the position as a **signed quantity** and let the sign do the work: a negative share count valued at the market price is a **negative position value**, which rises exactly when the price falls.

Note (Sign convention). A negative position value is a **liability**—a debt owed—not a loss already taken.

Note (Static versus constant-exposure shorts). The proof above holds the share count fixed at entry. A book that instead maintains a constant *dollar* exposure buys the short back down as it moves against it, so the unbounded loss is never realised. That daily rebalancing is an **implicit risk control**: the proof is right, but a constant-exposure book is not running the position it describes.

→ How this project implements it, with the accounting derivation and the measured gap between the two: `Backtests.md` in `Backtest_prototype/`.

1.4 Who Else Is Trading

Participant	Examples	Typical hold
Index / passive funds	Vanguard, BlackRock	Years to decades
Active managers	Fidelity, PIMCO	Monthly to quarterly
Hedge funds	—	Intraday to several days
Market makers	Optiver, Citadel Securities, IMC, SIG	Seconds to minutes
Noise traders	Retail, uninformed flow	No consistent horizon

Background

General finance knowledge needed to read this chapter, not part of its argument.

What is the relationship between futures/options and asset classes?

They answer different questions, so they are not alternatives to one another.

- **Asset class**—*what* underlying market you are exposed to.
- **Instrument**—*how* you hold that exposure: cash, futures, or options.

Futures are not an asset class, and options are not a rival one. Every asset class trades in every instrument:

Asset class	Typical underlying	Futures	Options
Equities	S&P 500, Nasdaq 100	ES, NQ	options on ES
Fixed income	US Treasuries, Bunds	ZN (10-year), ZB (30-year)	options on ZN
Commodities	Crude oil, gold, corn	CL, GC, ZC	options on CL
Currencies (FX)	EUR/USD, USD/JPY	6E, 6J	options on 6E

The same exposure comes in three wrappers: equity exposure is bought in cash as SPY, in futures as ES, or in options as an ES call. So in the opening definition—“trades **futures** systematically **across asset classes**”—the three parts are independent: **futures** is the wrapper, **across asset classes** is the breadth of underlying markets, and **systematically** means by rule or model rather than discretion.

Which asset class does a bond belong to?

Fixed income. Treasuries, corporate, municipal and high-yield bonds all sit there.

- “Fixed” describes the **contract**, meaning a scheduled coupon and the return of principal—not a fixed price.
- Bond prices still move, with interest rates, credit risk and time to maturity. That variation is what makes them a trend market at all.
- Treasury futures trade at 2-, 5-, 10- and 30-year maturities, so a CTA picks the *maturity* it wants exposure to.

Why do CTAs use futures rather than options?

	Futures	Options
Exposure	Linear —profit and loss move one-for-one with the underlying	Non-linear; depends on strike and moneyness
Extra drivers	None	Implied volatility, time decay, strike
What you are betting on	Direction	Direction and volatility

An option position is partly a volatility trade whether you intended it or not. Linear exposure is what lets one risk framework size 37 positions on the same scale.

Example 1.8. A trend follower might simultaneously hold long ES, short ZN, long GC, short 6J—all four are futures, and all four are different asset classes.

Appendix · Notation

Throughout, t is the date and i indexes an index’s constituents.

Symbol	Means	First used
I_t	the index level on date t —a published number, not a tradeable object	§1.1.1
P_{it}, N_i	constituent i ’s price on date t , and its float-adjusted share count	§1.1.1
M_t, D_t	the total float capitalisation, and the divisor that turns it into the printed level	§1.1.1
w_{it}	constituent i ’s share of total capitalisation—its weight in the index return	§1.1.1

Symbol	Means	First used
r_I, r_i	the index's return and constituent i 's return, over the same period	§1.1.1
s	the signed share count of a position: positive is long, negative is short	§1.3.3
P_0, P_T	the entry and exit price of that position	§1.3.3
$\text{PnL}(P_T)$	the position's profit and loss as a function of where the price ends up	§1.3.3
N	the number of shares borrowed to open a short	§1.3.2

Note (Collisions to watch). s here is a **signed share count**, not 2's s , which indexes the asset. w_{it} is a **constituent's weight inside an index**, fixed by capitalisation and not chosen by anyone; 2's $w_{s,t}$ and 5 · §3.3's w_i are **portfolio** weights, which are decisions. r_i is a constituent's return, whereas elsewhere in the course r carries an asset and a date subscript and means the forward return being predicted.